9.23 Solution: The electric and the magnetic fields of a TE mode is given by

$$\mathbf{E}_{lm}^{(M)} = Z_0 j_l(kr)\mathbf{L}Y_{lm}(\theta,\phi),$$

and

$$\mathbf{H}_{lm}^{(M)} = \mathbf{n}\frac{l(l+1)}{kr} j_l(kr)Y_{lm}(\theta,\phi) - \frac{i}{kr}\frac{\partial}{\partial r}\left[r j_l(kr)\right]\mathbf{n}\times\mathbf{L}Y_{lm}(\theta,\phi),$$

with $k = x_{ln}/a$, where x_{ln} is the n-th solution to

$$j_l(ka) = 0.$$

By Eq. (8.58), the power loss is

$$\begin{aligned} P_{\text{loss}} &= \frac{1}{2\sigma\delta}\int |\mathbf{n}\times\mathbf{H}|^2 da \\ &= \frac{1}{2\sigma\delta}\frac{1}{k^2}\left[\frac{\partial}{\partial r}\left[r j_l(kr)\right]\Big|_{r=a}\right]^2 \int |\mathbf{L}Y_{lm}(\theta,\phi)|^2 d\Omega \\ &= \frac{1}{2\sigma\delta}\frac{l(l+1)}{k^2}\left[\frac{\partial}{\partial r}\left[r j_l(kr)\right]\Big|_{r=a}\right]^2. \end{aligned}$$

The total energy in the cavity is

$$U = \frac{\varepsilon_0}{2}\int |\mathbf{E}|^2 d^3x = \frac{\mu_0}{2}\int_0^a r^2 j_l(kr)^2 dr \int |\mathbf{L}Y_{lm}(\theta,\phi)|^2 d\Omega = \frac{\mu_0}{2} l(l+1)\int_0^a r^2 j_l(kr)^2 dr.$$

The integral can be performed by using the definition of the spherical Bessel function,

$$j_l(r) = \sqrt{\frac{\pi}{2r}} J_{l+1/2}(r),$$

which gives

$$\int_0^a r^2 j_l(kr)^2 dr = \frac{\pi}{2k}\int_0^a r J_{l+1/2}\left(x_{ln}\frac{r}{a}\right)^2 dr = \frac{\pi}{2k}\frac{a^2}{2} J_{l+3/2}(ka)^2 = \frac{a^3}{2} j_{l+1}(ka)^2.$$

Also,

$$\frac{\partial}{\partial r}\left[r j_l(kr)\right]\Big|_{r=a} = j_l(ka) + ka j_l'(ka) = ka j_{l+1}(ka).$$

Therefore,

$$Q = \omega\frac{U}{P_{\text{loss}}} = \omega\mu_0\sigma\delta\frac{a}{2}.$$

Since

$$\delta = \sqrt{\frac{2}{\mu_0\omega\delta}},$$

then

$$Q = a\sqrt{\frac{\mu_0\omega\delta}{2}} = \frac{a}{\delta},$$

for all TE modes.

For TM modes,

$$\mathbf{H}_{lm}^{(E)} = j_l(k'r)\mathbf{L}Y_{lm}(\theta,\phi),$$

and

$$\mathbf{E}_{lm}^{(E)} = -\mathbf{n}\frac{Z_0 l(l+1)}{k'r} j_l(k'r)Y_{lm}(\theta,\phi) + \frac{iZ_0}{k'r}\frac{\partial}{\partial r}\left[r j_l(k'r)\right]\mathbf{n}\times\mathbf{L}Y_{lm}(\theta,\phi),$$

with $k' = x'_{ln}/a$, where x'_{ln} is the n-th solution to

$$\frac{d}{dx}\left[x j_l(x)\right] = 0.$$

Similarly, we can calculate the power loss as

$$P_{\text{loss}} = \frac{1}{2\sigma\delta}\int |\mathbf{n}\times\mathbf{H}|^2 dS = \frac{a^2}{2\sigma\delta} l(l+1) j_l(k'a)^2,$$

and the energy

$$U = \frac{\mu_0}{2}\int |\mathbf{H}|^2 d^3x = \frac{\mu_0}{2} l(l+1)\int_0^a r^2 j_l(k'r)^2 dr.$$

Using the integration identity

$$\int_0^a r^2 j_l(k'r)^2 dr = \frac{r^3}{2}\left(j_l(k'r)^2 - j_{l-1}(k'r)j_{l+1}(k'r)\right) + C,$$

the energy becomes

$$U = \frac{\mu_0}{2} l(l+1)\frac{a^3}{2}\left(j_l(k'a)^2 - j_{l-1}(k'a)j_{l+1}(k'a)\right).$$

To simplify the energy expression, form the recursion relations Eq. (9.90), we have

$$\frac{d}{dx}\left[x j_l(x)\right] = x j_{l-1}(x) - l j_l(x).$$

Let $x = k'a$, by the equations of the characteristic equations, the left hand side is 0. Therefore,

$$j_{l-1}(k'a) = \frac{l}{k'a} j_l(k'a).$$

Then, using

$$\frac{2l+1}{x} j_l(x) = j_{l+1}(x) + j_{l-1}(x),$$

we have

$$j_{l+1}(k'a) = \frac{2l+1}{k'a} j_l(k'a) - j_{l-1}(k'a) = \frac{2l+1}{k'a} j_l(k'a) - \frac{l}{k'a} j_l(k'a) = \frac{l+1}{k'a} j_l(k'a).$$

Therefore,

$$U = \frac{\mu_0}{2} l(l+1)\frac{a^3}{2} j_l(k'a)^2\left(1 - \frac{l(l+1)}{(k'a)^2}\right),$$

and

$$Q = \omega\frac{U}{P_{\textbf{loss}}} = \frac{a}{\delta}\left(1 - \frac{l(l+1)}{{x'_{ln}}^2}\right).$$